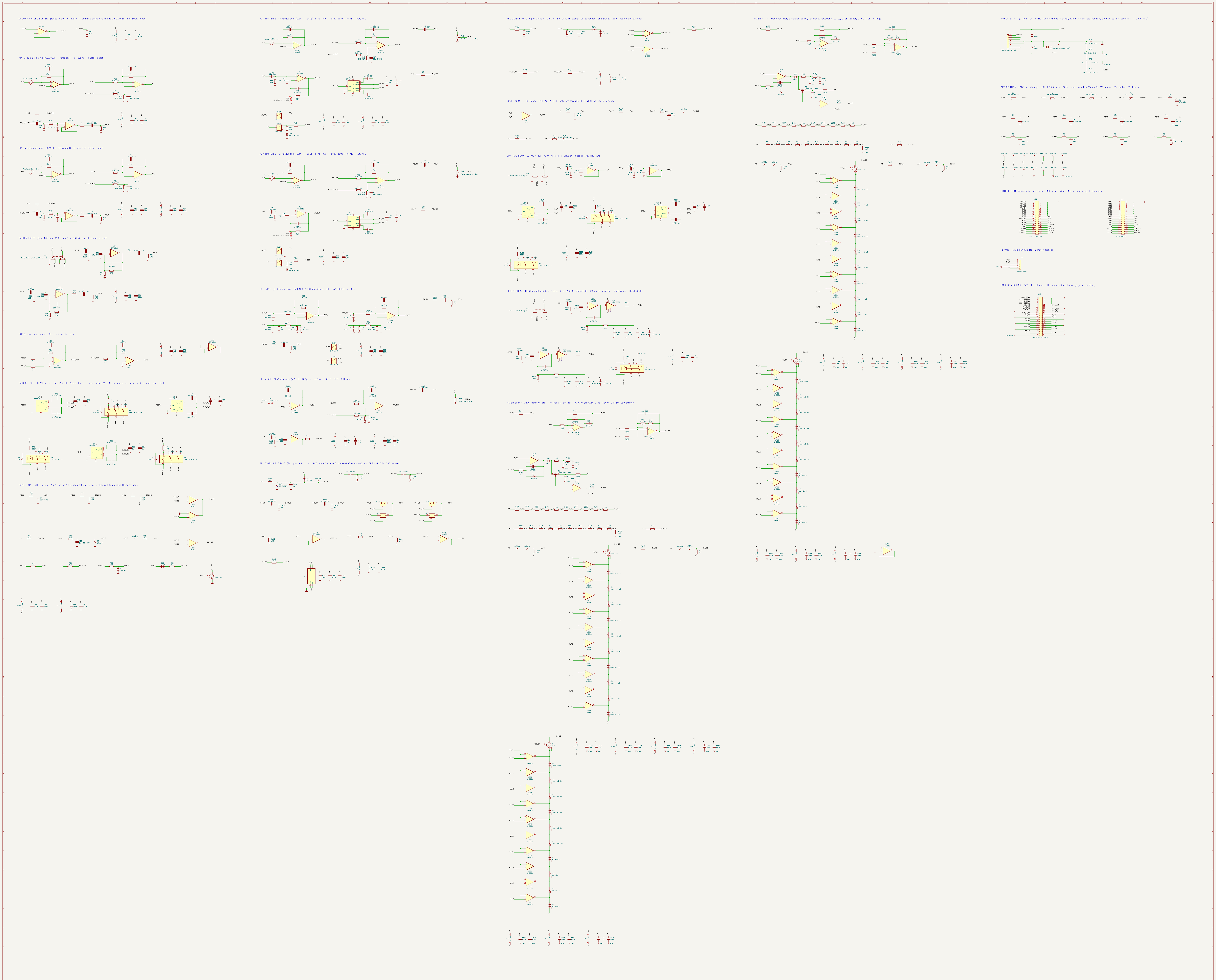

GROUND CANCEL BUFFER (feeds every re–inverter; summing amps use the raw GCANCEL line; 100K keeper)
MIX L: summing amp (GCANCEL–referenced), re–inverter, master insert
MIX R: summing amp (GCANCEL–referenced), re–inverter, master insert
MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post–amps +10 dB
MONO: inverting sum of POST L+R, re–inverter
MAIN OUTPUTS: DRV134 –> 10u NP in the Sense loop –> mute relay (NO; NC grounds the line) –> XLR male, pin 2 hot
POWER–ON MUTE: rails > ~14 V for ~2.7 s closes all six relays; either rail low opens them at once
AUX MASTER 5: OPA1612 sum (22K || 100p) + re–invert, level, buffer, DRV134 out, AFL
AUX MASTER 6: OPA1612 sum (22K || 100p) + re–invert, level, buffer, DRV134 out, AFL
EXT INPUT (2–track / DAW) and MIX / EXT monitor select (SW latched = EXT)
PFL / AFL: OPA1656 sum (22K || 100p) + re–invert, SOLO LEVEL, follower
PFL SWITCHER: DG413 (PFL pressed = SW1/SW4, else SW2/SW3; break–before–make) –> CRS L/R OPA1656 followers
PFL DETECT (0.92 V per press vs 0.50 V; 2 x 1N4148 clamp, 1u debounce) and DG413 logic, beside the switcher
RUDE SOLO: ~2 Hz flasher, PFL ACTIVE LED; held off through FL_N while no key is pressed
CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs
HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, mute relay, PHONESGND
METER L: full–wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10–LED strings
METER R: full–wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10–LED strings
POWER ENTRY (7–pin XLR NC7MD–LX on the rear panel, two 5 A contacts per rail, 18 AWG to this terminal; +–17 V PSU)
DISTRIBUTION (PTC per wing per rail, 1.85 A hold, 72 V; local branches VA audio, VP phones, VM meters, VL logic)
MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)
REMOTE METER HEADER (for a meter bridge)
JACK BOARD LINK 2x20 IDC ribbon to the master jack board (9 jacks, 3 XLRs)
RV1 Master fader 10K log 100mm dual
RV5 C/Room level 10K log dual
RV6 Phones level 10K log dual
J1 PSU in (NC7MD–LX)
CN1 Ground bar M4 (star point)
CN2 Bus L wing 2x17
CN3 Bus R wing 2x17
CN4 Remote meter
CN5 Jack board link 2x20

Drawn by the master_lib fragments (tools/frags.py); refs auto–assigned
Generated by tools/gen_master.py from master_section_spec.md v0.9

Sheet: /
File: master.kicad_sch

Title: Master section — goa console

Size: User	Date: 2026–10–19	**Rev: 0.9**
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